DAYANANDA SAGAR
College of Arts, Science & Commerce



PG – 505

II Semester M.C.A. Degree Examination, Oct./Nov. 2020
(CBCS)
COMPUTER SCIENCE
MCA 203 T : Computer Networks

Time : 3 Hours | Max. Marks : 70

Instruction : *Answer **any five** questions from Section – **A** and answer **any four** questions from Section – **B**.*

SECTION – A

Answer **any five** questions from eight, **each** carries **six** marks. **(5×6=30)**

1. Discuss working of coaxial cables and optical fibre cables with the help of diagram. **6**

2. a) Define bandwidth. How it relates to channel efficiency ?

   b) 1000 baud channel with 6 levels is used for communication. What will be the maximum speed of the channel ? **(3+3)**

3. Explain working of Ring topology with suitable example. How ring topology sustain physical failure ? **6**

4. Clarify UP stream, down stream terminology in communication. Why we require different streaming in cable modem technology for networking ? **6**

5. Explain message communication through ISO-OSI architecture. **6**

6. Explain IPV4 addressing format. Give examples for Class A, Class B, Class C address. **6**

7. What is the use of DNS ? How it works in internet ? **6**

8. Explain MVC design pattern. **6**

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SECTION – B

Answer **any four** questions, **each** carries **ten** marks. **(4×10=40)**

9. A) How do you detect transmission error ? With an example explain CRC technique.

   B) Consider 110011 as data and 1101 as generator. Show how various data transmission if CRC is following. **10**

10. What is MAC address ? Does it has some link or something in common to Mac OS of Apple ? Explain MAC frame format and addressing system. **(7+3)**

11. List and explain any five different network interface hardware devices used in computer networking. Explain their performance layers in 7 layers architecture. **10**

12. What is routing ? Explain link state routing protocol with suitable example. **10**

13. Explain E-mail format. How mail transfer and POP happens for messaging ? **10**

14. Explain the purpose of ARP in message transfer. Show step by step procedure in ARP resolution. **10**

15. Write short notes on VOIP, ICMP, WWW. **10**

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II Semester M.C.A. Examination, Nov./Dec. 2022

(CBCS Scheme) (2020-21 and Onwards)

COMPUTER SCIENCE

2MCA3 : Computer Networks

Time : 3 Hours Max. Marks : 70

Instructions : *Part – **A** : Answer **any five** questions.*

*Part – **B** : Answer **any four** questions.*

PART – A

Answer **any five** questions. **Each** question carries **six** marks : **(5×6=30)**

1. Explain half and full duplex communication in brief with example.
2. Define network topology. Explain different topologies with its merits and demerits.
3. What is switching ? Explain circuit switching technique.
4. Define multiplexing. Explain its types.
5. Discuss the functioning of ISO-OSI layering model.
6. What do you mean by transmission error ? With an example explain Cyclic Redundancy Check Technique.
7. Explain ALOHA and slotted ALOHA.
8. What are bridges ? Explain its types with advantages and disadvantages.

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PART – B

Answer **any four** questions. **Each** question carries **ten** marks : **(4×10=40)**

9. Explain TCP/IP Protocol suite with a neat diagram. **10**

10. Explain in brief the different types of guided transmission media. **10**

11. What is IPv4 addressing format ? Explain class A, class B and class C addressing format with suitable examples. **10**

12. Write a note on :

    a) Point to Point (PPP)

    b) HDLC. **(4+6)**

13. a) Explain FDMA, TDMA and CDMA. **6**

    b) Write a note on Domain Name System. **4**

14. a) Write a note on 802.11 wireless architecture. **5**

    b) Write a short note on congestion control methods. **5**

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II Semester M.C.A. Examination, Nov./Dec. 2023
(CBCS) (2020-21 and Onwards)
COMPUTER SCIENCE
2MCA3 : Computer Networks

Time : 3 Hours Max. Marks : 70

Instructions : *1) Part – **A** : Answer **any five** questions.*
*2) Part – **B** : Answer **any four** questions.*

PART – A

Answer **any five** questions. **Each** question carries **six** marks. **(5×6=30)**

1. Define network topology. Explain different topologies with their merits and demerits.
2. What is switching ? Explain message switching technique.
3. Write a note on Transmission Impairment.
4. Define multiplexing. Explain FDM and WDM.
5. Discuss the TCP/IP architecture.
6. What do you mean by Burst error ? With an example explain Checksum technique.
7. Explain Carrier Sense Multiple Access (CSMA) with its methods.
8. Give a note on Mobile IP.

PART – B

Answer **any four** questions. **(4×10=40)**

9. Explain ISO-OSI reference model with a neat diagram. **10**
10. Explain in brief the different types of Guided and Unguided transmission media. **10**
11. Explain in detail the HDLC protocol with its Frame format. **10**

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12. a) Explain Reservation and Polling controlled access methods. 6

    b) Write a note on Unicast, Multicast and Broadcast address. 4

13. a) Explain IP Address with IP packet structure. 6

    b) Write a note on IPV6 address. 4

14. a) Explain the features of Transmission Control Protocol (TCP). 6

    b) Write a short note on Link State Routing Algorithm. 4

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II Semester M.C.A. Examination, December 2024
(CBCS) (2020–21 and Onwards)
COMPUTER SCIENCE
2MCA3 : Computer Networks

Time : 3 Hours Max. Marks : 70

Instruction : *Answer **any five** from Section – **A** and **any four** from Section – **B**.*

SECTION – A

Answer **any five** of the following. **Each** question carries **six** marks : **(5×6=30)**

1. Define computer networks. Explain its characteristics.
2. Explain the various transmission modes used in network.
3. With a neat diagram, explain Pure and Slotted ALOHA.
4. What is switching and explain packet switching.
5. Explain the purpose of each layer in the TCP/IP reference model.
6. Explain Cyclic Redundancy Check (CRC) and check sum technique with example.
7. Differentiate between TCP and UDP.
8. Define mobile IP and explain how it works.

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SECTION – B

Answer **any four** of the following. **Each** question carries **ten** marks : **(4×10=40)**

9. Explain CSMA/CD and CSMA/CA technique in detail.
10. a) Write a short note on : 6
    DNS, PPP, HDLC
    b) What is QoS ? What are the importance of Quality of Services (QoS) ? 4

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11. a) Define network topology. Explain BUS and STAR topologies with merits and demerits. 6

    b) Differentiate between link state and distance vector routing algorithm. 4

12. a) Explain reservation and polling controlled access methods. 6

    b) Write a note on IPV6 address. 4

13. Explain ISO-OSI reference model with a neat diagram. 10

14. a) What is IPV4 addressing format ? Explain class A, class B and class C addressing format. Identify the following address. 7

    a → 192.10.127.35 b → 14.12.22.3

    b) Explain the working of SMTP. 3

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II Semester M.C.A. Degree Examination, November/December 2025
(CBCS) (2020-21 and Onwards)
COMPUTER SCIENCE
2MCA3 : Computer Networks

Time : 3 Hours | Max. Marks : 70

***Instruction** : Answer **all** the Sections.*

SECTION – A

I. Answer **any five** questions of the following. **Each** question carries **6** marks. **(5×6=30)**

1) Define computer network. Explain the different transmission modes used in computer networks.

2) Explain different types of topologies. And assume that six devices are arranged in a Mesh Topology. How many cables are needed ? How many ports are needed for each device ?

3) Briefly explain Cyclic Redundancy Check technique (CRC). Find the CRC for this given message M = 1010001101 using the divisor polynomial $x^5 + x^4 + x^2 + 1$.

4) Briefly explain HDLC and PPP protocols.

5) What is Transmission Error ? Explain checksum with an example.

6) What is IP addressing ? What are the types of IPV4 classification ? A network on the Internet has a subnet mask of 255.255.240.0. What is the maximum number of hosts it can handle ?

7) Explain the Distance Vector Routing algorithm with an example.

8) Briefly explain the TCP/IP Protocol.

SECTION – B

II. Answer **any four** questions. **Each** question carries **ten** marks. **(4×10=40)**

9) Explain the working of the physical layer. Determine the maximum length of the cable (in km) for transmitting data at a rate of 500 Mbps in an Ethernet LAN with frames of size 10000 bits. Assume the signal speed in the cable to be 200000 km/s. (1Mbps = 10^6 bits per second). **10**

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10) a) Differentiate between ALOHA and Slotted ALOHA. 5

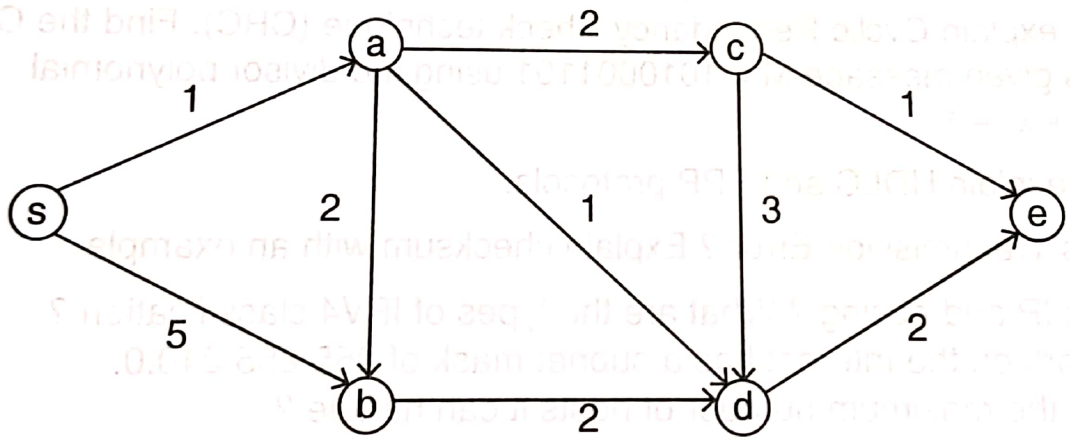
b) A group of N stations share a 100 Kbps slotted ALOHA channel. Each station outputs a 500-bit frame on average of 5000 ms, even if the previous one has not been sent. What is the required value of N ? 5

11) Briefly explain the TCP segment format. The following is a dump of a TCP header in hexadecimal format. Find the source port number, destination port number, sequence number, acknowledgement number, length of the header and type of the segment and what is the window size ?
TCP header is : E293 0017 00000001 00000000 5002 07FF..... 10

12) a) What are the differences between classful addressing and classless addressing in IPV4 ? 6

b) Write a note on Domain Name Server. 4

13) a) Explain Dijkstra's Algorithm. Using Dijkstra's Algorithm, find the shortest distance from source vertex 'S' to remaining vertices in the following graph. Also, write the order in which the vertices are visited. 6



b) Write a short note on congestion control methods. 4

14) a) What is firewall used for ? How network security can be managed from attacks ? 5

b) Write a note on Unicast Routing protocols. 5

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